Permutations, Inversions, and Determinants

N-20220725

§1.1 Inversions and signs

For a permutation
$$\sigma = (\sigma_1, \dots, \sigma_n),$$
an inversion is a pair $i < j$ such that
$$\sigma_i > \sigma_j.$$
The number of inversions is denoted $\tau(\sigma)$. The sign of the permutation is
$$\operatorname{sgn}(\sigma) = (-1)^{\tau(\sigma)}.$$
For example, for 426315, the inversions are counted by checking how many later entries are smaller than each entry. The note obtains
$$\tau(426315) = 8.$$

In determinant expansions, the sign of each term is exactly the sign of a permutation. Thus a term like
$$a_{21}a_{53}a_{16}a_{42}a_{65}a_{34}$$
corresponds to a row/column permutation, and its sign is determined by its inversion number.

§1.2 Definition of determinant

Definition 1.2.1 (Determinant)**.** The determinant is the alternating multilinear function of the columns of a square matrix normalized by
$$\det I = 1.$$
It measures oriented volume scaling.

The determinant of an $n \times n$ matrix $A = (a_{ij})$ is
$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)}.$$
This formula is not meant for computation in large dimensions. Its value is conceptual: it explains why row swaps change sign, why repeated rows give zero, and why the determinant is multilinear in rows and columns.

§1.3 Basic determinant properties

The standard properties are:

(i) Transposition does not change the determinant:

$$\det A^T = \det A.$$

(ii) Swapping two rows changes the sign.

(iii) Pulling a scalar out of one row pulls it out of the determinant.

(iv) The determinant is additive in each row separately.

(v) Adding a multiple of one row to another row does not change the determinant.

From these one immediately gets the zero properties:

(i) if a row is zero, then the determinant is zero;

(ii) if two rows are proportional, then the determinant is zero.

§1.4 Sarrus and triangular matrices

For a 3×3 matrix, Sarrus' rule gives a convenient visual computation. But the note also emphasizes that triangular matrices are easier: if A is upper or lower triangular, then

$$\det A = a_{11}a_{22}\cdots a_{nn}.$$

This is why row reduction is so powerful. We use row operations to simplify a determinant into triangular form while tracking sign changes and scalar factors.

§1.5 Cofactor expansion

The cofactor of a_{ij} is

$$C_{ij} = (-1)^{i+j} M_{ij},$$

where M_{ij} is the determinant obtained by deleting row i and column j. Expanding along row i,

$$\det A = \sum_{j=1}^{n} a_{ij} C_{ij}.$$

A page in the note works through determinant computations by choosing rows or columns with many zeros. This is the main practical advice: do not expand randomly; expand where the determinant is sparse.

§1.6 A structured determinant

A common determinant in the note has the form

$$D_n = \begin{vmatrix} x & a & a & \cdots & a \\ a & x & a & \cdots & a \\ \vdots & \vdots & \vdots & & \vdots \\ a & a & a & \cdots & x \end{vmatrix}.$$

This matrix can be written

$$D_n = (x - a)I + aJ,$$

where J is the all-one matrix. The vector $(1, \dots, 1)$ is an eigenvector with eigenvalue

$$x + (n - 1)a,$$

and any vector whose coordinates sum to 0 has eigenvalue

$$x - a.$$

Therefore

$$D_n = (x + (n - 1)a)(x - a)^{n-1}.$$

The note obtains this by row and column operations; the eigenvalue argument is the higher-level explanation.

§1.7 The determinant line and exterior powers

The previous computational rules all point to one structure. Let V be an n-dimensional vector space. The top exterior power

$$\Lambda^n V$$

is one-dimensional and is called the determinant line. A nonzero element of this line is a volume form; a choice up to positive scalar is an orientation.

Definition 1.7.1 (Determinant via the top exterior power)**.** For a linear map $A : V \to V$, the induced map

$$\Lambda^n A : \Lambda^n V \to \Lambda^n V$$

acts on a one-dimensional space, hence is multiplication by a scalar. That scalar is $\det A$.

In a basis $e_1, \dots, e_n$, if the columns of A are $v_1, \dots, v_n$, then

$$v_1 \wedge \cdots \wedge v_n = (\det A) e_1 \wedge \cdots \wedge e_n.$$

This single identity explains the main determinant rules:

(i) multilinearity comes from multilinearity of the wedge product;

(ii) repeated or dependent columns wedge to zero;

(iii) swapping two columns changes the sign;

(iv) multiplicativity follows from $\Lambda^n(AB) = (\Lambda^n A)(\Lambda^n B)$.

§1.8 Why permutation signs are forced

The sign of a permutation is not an arbitrary convention. It is the unique homomorphism

$$S_n \longrightarrow \{\pm 1\}$$

that sends every transposition to -1. The inversion number gives a concrete formula:

$$\operatorname{sgn}(\sigma) = (-1)^{\#\text{ inversions of } \sigma}.$$

Because the determinant is alternating, permuting the columns by σ must multiply the volume form by this sign. This is why the Leibniz formula has exactly the sign pattern it has.

§1.9 Minors as exterior coordinates

The same exterior-power viewpoint also explains minors. If $A : V \to W$, then

$$\Lambda^k A : \Lambda^k V \to \Lambda^k W$$

is represented, in the induced exterior bases, by the $k \times k$ minors of A. Therefore

$$\operatorname{rank} A < k \quad \Longleftrightarrow \quad \Lambda^k A = 0 \quad \Longleftrightarrow \quad \text{all } k \times k \text{ minors vanish.}$$

Thus minors are not arbitrary subdeterminants; they are coordinates of an exterior-power map.

§1.10 Cramer's rule from volume ratios

Cramer's rule replaces one column of A by b. Geometrically, solving $Ax = b$ asks for the coordinates of b in the parallelepiped basis given by the columns of A. The coordinate x_i is the ratio of two oriented volumes:

$$x_i = \frac{\det A_i(b)}{\det A}.$$

This interpretation makes the formula less mysterious: a coordinate is measured by replacing the corresponding basis vector and comparing volumes.